

# Lineage Before Holonomy

A certificate stack for representation identity under context, checkpoint, and precision change

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[Repository and feature branch](#) · [Focused Qwen precision report](#)

**Evidence boundary.** The mathematical instruments are validated on exact and CPU-synthetic fixtures. The real-model program supports a narrow context-conditioned rank-one lineage result and several negative or unavailable outcomes. It has not established a noise-valid higher-rank transformer holonomy signal, causal transformer edit value, semantic identity, monitor sufficiency, capability preservation, recursive improvement, or deployment safety.

## Abstract

Representation comparisons usually ask whether two activation sets span similar subspaces. That question is necessary but insufficient when a signed monitor or edit coordinate must retain one identity across several context, checkpoint, site, or precision changes. We organize this problem as a certificate stack. A local spectral object must first clear a support gate; adjacent conditions must then share a common-rank lineage; the admitted graph must contain cycles; polar transports may then be composed into loop holonomy; and any categorical holonomy result must pass a liveness or matched-noise gate before it can enter a causal or edit-sectioning test. The ordering is logical—a partial order of certificate availability—rather than a universal temporal law of representation degradation.

Synthetic controls establish why the layers are distinct. Two rank-two systems have the same minimum edge retention, 0.990033, yet the curved system accumulates  $62.173^\circ$  of loop rotation and returns only 0.466809 signed alignment while gauge-invariant block energy remains exactly one. Planted curvature scales with loop area ( $R^2 = 0.999975$ ), whereas flat-connection estimator-noise phase variance scales with perimeter ( $R^2 = 0.947972$ ). On a planted causal fixture, frozen holonomy features reduce grouped held-out squared error by 89.0005% and a conservative bound authorizes 91.291% of cases with zero false authorizations. A separate sectioning experiment shows that holonomy patches help only when the patch boundary matches edit identity.

The transformer evidence is more restrictive. An initial Qwen0.8B local construction supports no rank at any of 512 nodes, although pooling contexts recovers 60 of 64 global objects. A denser capture recovers one rank-one graph-reachability object across four layers; checkpoint edges are nearly unit-retention, while context edges determine every connectivity threshold. An independent Qwen1.7B study supports zero of 12 global module identities, with prompt identity explaining more than 95% of the surviving retention variation. Finally, a paired precision study shows that 4-bit quantization compresses all eight layer-19 support margins and erodes a graded loop margin, but its positive rank-one holonomy signs are not evidence: if  $r_e$  is edge retention and  $\alpha_e = \arccos \sqrt{r_e}$ , negative rank-one holonomy requires  $\sum_{e \in C} \alpha_e \geq \pi$ . Every measured cycle lies below that bound. The durable conclusion is therefore *lineage before holonomy*: context-conditioned lineage is presently the binding real-model layer, and holonomy remains a downstream audit whose liveness must be demonstrated.

# 1 Introduction

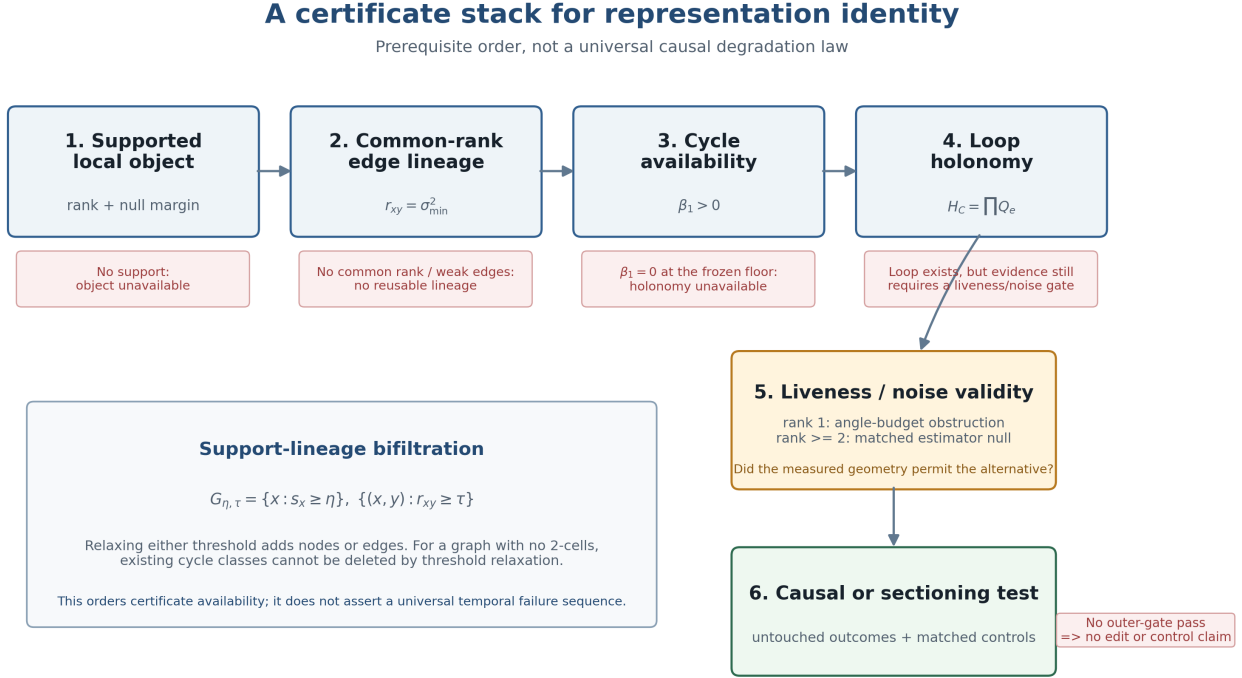
A representation feature is often treated as the same object after a change of prompt, checkpoint, fine-tuning state, model site, or runtime precision. Standard similarity measures such as SVCCA and CKA compare activation sets up to useful classes of transformation [Raghu et al., 2017, Kornblith et al., 2019]. Model stitching adds a functional test by asking whether representations can be connected without large task loss [Bansal et al., 2021]. These tools answer important questions, but they do not by themselves certify that a *signed coordinate* retains one path-independent identity across several transformations. Similarity metrics can disagree or become insensitive to behaviorally consequential changes, motivating explicit functional and statistical grounding [Ding et al., 2021].

The missing object is composition. Suppose a representation subspace is measured at four corners of a context-by-checkpoint square. Each adjacent pair can have excellent principal-angle retention, yet the best orthogonal transports can accumulate a nontrivial rotation around the loop. Local overlap and global coordinate consistency are therefore different observables. Holonomy is the loop product of those transports. A recent ICLR 2026 paper independently develops gauge-invariant representation holonomy on input-space loops using globally whitened local feature clouds and rotation-only Procrustes alignment, and relates it to robustness [Sevetlidis and Pavlidis, 2026]. The present work studies a complementary setting: condition-indexed, target-blind spectral subspaces over discrete context/checkpoint/precision graphs; explicit support, common-rank, cycle, orientation, noise, and liveness gates; the full orthogonal group  $O(r)$  rather than an  $SO(r)$ -only estimator; and a sequence dominated by real-model negative results rather than a robustness correlation.

The paper’s contribution is not the claim that holonomy already controls language models. It is an evidence hierarchy showing when a holonomy statement is even available and when it carries information beyond the metric geometry that generated it. The main contributions are:

- C1. A certificate stack.** We separate local spectral support, common-rank lineage, graph-cycle availability, loop holonomy, liveness/noise validity, and outer causal or sectioning value. The separation prevents an unavailable object, a tree, a forced sign, or a noise-dominated loop from being narrated as a scientific null.
- C2. A support-lineage bifiltration.** Two frozen thresholds generate a monotone family of condition graphs. Existing graph-cycle classes persist as either threshold is relaxed. This supplies a legitimate ordering of certificate availability without claiming a universal chronological failure sequence.
- C3. Synthetic separation and causal calibration.** Matched local lineage can conceal large closed-loop signed-coordinate loss; area/perimeter scaling separates planted curvature from estimator noise; and a planted causal fixture shows what a successful downstream gate would look like.
- C4. A rank-one liveness theorem.** Negative line-bundle holonomy requires total projective principal-angle budget at least  $\pi$ . Below that bound, a positive sign is forced and adds no evidence beyond edge lineage.
- C5. A convergent transformer result.** Across Qwen0.8B and Qwen1.7B captures, context variation repeatedly limits common-rank lineage more strongly than checkpoint variation. Runtime quantization further compresses support and loop margins before discrete topology becomes live.

Figure 1 summarizes the stack. It should be read from left to right as a sequence of prerequisites for stronger claims, not as an assertion that all representations fail in that physical order.



**Figure 1:** The proposed certificate stack. A representation holonomy claim presupposes a supported local object, a common-rank lineage graph, and a nontrivial cycle space. A categorical loop result then requires a liveness or matched-noise gate, and no operational claim follows without an untouched causal or sectioning test. The lower-left panel is the two-threshold support-lineage graph family.

## 2 Mathematical framework

### 2.1 Condition-indexed spectral objects

Let  $X$  be a finite set of measurement conditions. A condition may include a prompt family or shard, a checkpoint or adapter state, a model site, a runtime precision, and a replicate identifier. At each  $x \in X$ , a frozen target-blind construction either returns an  $r_x$ -dimensional subspace

$$E_x \in \text{Gr}(r_x, d)$$

or declares the object unsupported. The exact object differs across the experiments below—between-class directions, reliability-weighted spectral bands, or VPD-derived families—and statistics are never pooled across nonidentical constructions.

Choose an orthonormal frame  $B_x \in \mathbb{R}^{d \times r}$  only after a common rank  $r$  has been admitted on the relevant graph. The frame is a coordinate choice, not the object. A gauge change is

$$B_x \mapsto B_x G_x, \quad G_x \in \text{O}(r).$$

Each node carries a signed support margin  $s_x$  measuring distance from a registered null-plus-margin boundary. The implication is deliberately one-sided:

$$s_x \geq \eta \implies E_x \text{ is supported at threshold } \eta.$$

Failure does not prove that the representation vanished; it says that this experiment cannot certify the object at that node.

## 2.2 Edge lineage and polar transport

For adjacent supported nodes  $x$  and  $y$ , define the overlap

$$M_{y \leftarrow x} = B_y^\top B_x.$$

If

$$M_{y \leftarrow x} = U \Sigma V^\top$$

is an SVD, its singular values are the cosines of principal angles between  $E_x$  and  $E_y$  [Björck and Golub, 1973, Edelman et al., 1998, Stewart and Sun, 1990]. We use the worst-direction retention

$$r_{xy} = \sigma_{\min}(M_{y \leftarrow x})^2 = \cos^2 \theta_{\max}(E_x, E_y). \quad (1)$$

The orthogonal polar factor

$$Q_{y \leftarrow x} = UV^\top \in O(r) \quad (2)$$

is the closest norm-preserving coordinate transport in the ordinary Procrustes sense [Schönemann, 1966]. The singular values and  $Q$  play different roles:  $\Sigma$  measures local subspace retention, while  $Q$  carries signed-coordinate transport. Retention-only statistics intentionally discard  $Q$ .

Under independent gauge changes at the endpoints,

$$Q_{y \leftarrow x} \mapsto G_y^\top Q_{y \leftarrow x} G_x. \quad (3)$$

Thus individual matrix entries are gauge-dependent, while closed-loop conjugacy data are not. This construction is a finite graph analogue of connection-based transport methods such as vector diffusion maps [Singer and Wu, 2012].

## 2.3 The support-lineage bifiltration

For a support threshold  $\eta$  and lineage threshold  $\tau$ , define

$$V_\eta = \{x \in X : s_x \geq \eta\}, \quad (4)$$

$$E_{\eta, \tau} = \{(x, y) : x, y \in V_\eta, r_{xy} \geq \tau\}, \quad (5)$$

$$G_{\eta, \tau} = (V_\eta, E_{\eta, \tau}). \quad (6)$$

This is a two-parameter filtration in the elementary graph-theoretic sense. It is related in spirit to multiparameter persistence [Carlsson and Zomorodian, 2009], but the present work does not claim a complete multipersistence invariant or classification.

**Proposition 1** (Monotone availability and cycle persistence). *If  $\eta' \leq \eta$  and  $\tau' \leq \tau$ , then*

$$G_{\eta, \tau} \subseteq G_{\eta', \tau'}.$$

*For coefficients in any field, the induced map*

$$H_1(G_{\eta, \tau}) \longrightarrow H_1(G_{\eta', \tau'})$$

*is injective. Consequently, relaxing support or lineage thresholds cannot delete an existing graph-cycle class, and  $\beta_1$  is nondecreasing along every threshold-relaxation path.*

*Proof.* The inequalities enlarge the eligible vertex and edge sets, giving a subgraph inclusion. A graph is a one-dimensional CW complex, so its cellular chain complex has  $C_2 = 0$  and therefore  $B_1 = \text{im } \partial_2 = 0$ . A nonzero cycle in the subgraph remains a nonzero one-chain in the supergraph and cannot become a boundary there. Hence the induced  $H_1$  map is injective.  $\square$

The proposition legitimizes a sorting pattern: support precedes edge lineage as a certificate requirement, and cycle availability precedes holonomy. It does *not* impose a universal relationship between first connectivity and first cycle appearance. A cycle can close inside one component before the whole graph connects, or a connected tree can appear before any cycle.

For a descending edge filtration, we record

$$\begin{aligned}\tau_{\text{conn}} &= \sup\{\tau : G_{\eta,\tau} \text{ is connected}\}, \\ \tau_{\text{cycle}} &= \sup\{\tau : \beta_1(G_{\eta,\tau}) > 0\}.\end{aligned}$$

A frozen floor may make holonomy unavailable even when cycles exist at a slightly lower threshold. This distinction becomes important in the Qwen0.8B layer-15 reanalysis.

## 2.4 Loop holonomy and gauge invariance

For an oriented cycle  $C = (x_0, x_1, \dots, x_m = x_0)$ , use one frozen multiplication convention and define

$$H_C = Q_{x_0 \leftarrow x_{m-1}} \cdots Q_{x_2 \leftarrow x_1} Q_{x_1 \leftarrow x_0} \in O(r). \quad (7)$$

Equation (3) gives

$$H_C \mapsto G_{x_0}^\top H_C G_{x_0}.$$

Therefore  $\det H_C$ ,  $\text{tr } H_C$ , and the sorted canonical rotation-plane angles of its orientation-preserving component are gauge-invariant. In rank one,

$$H_C \in O(1) = \{+1, -1\}.$$

The assignment of negative signs to cycle classes is the first Stiefel–Whitney class  $w_1$  of the induced real line bundle [Milnor and Stasheff, 1974]; a sign +1 means  $w_1(C) = 0$ , not “ $w_1 = +1$ .”

For  $r \geq 2$ , orientation is a prior discrete gate. If  $\det H_C < 0$ , a reflection has occurred and rotation-angle summaries are not a substitute. If  $\det H_C > 0$ , canonical angles summarize the conjugacy class in  $SO(r)$ . In either case, a real-model loop requires an estimator-noise null matched on rank, perimeter, retention, and uncertainty before geometric curvature is inferred.

## 2.5 Rank-one liveness

Rank one has no continuous  $SO(1)$  angle, but its discrete sign is constrained by the same edge geometry that generated it.

**Theorem 2** (Rank-one holonomy liveness). *Let  $C = (\ell_0, \ell_1, \dots, \ell_m = \ell_0)$  be a cycle of real lines. Let  $\alpha_i \in [0, \pi/2]$  be the principal angle between  $\ell_i$  and  $\ell_{i+1}$ . Under shortest-geodesic rank-one transport, negative holonomy implies*

$$\sum_{i=0}^{m-1} \alpha_i \geq \pi. \quad (8)$$

*Consequently, if the total angle budget is below  $\pi$ , the holonomy sign is forced to be +1.*

*Proof.* Choose a unit lift  $v_0 \in \ell_0$ . Recursively choose  $v_{i+1} \in \ell_{i+1}$  with  $v_i^\top v_{i+1} \geq 0$ . The spherical geodesic distance between successive lifts is  $\alpha_i$ . Since  $\ell_m = \ell_0$ , the terminal lift is  $v_m = \sigma_C v_0$  for  $\sigma_C \in \{+1, -1\}$ . If  $\sigma_C = -1$ , the path joins antipodes, whose spherical distance is  $\pi$ . The triangle inequality gives

$$\pi = d(v_0, -v_0) \leq \sum_i d(v_i, v_{i+1}) = \sum_i \alpha_i.$$

□

With rank-one retention  $r_e$ , Equation (1) gives

$$\alpha_e = \arccos \sqrt{r_e}. \quad (9)$$

Define the cycle and graph frustration margins

$$F(C) = \pi - \sum_{e \in C} \alpha_e, \quad (10)$$

$$F(G) = \pi - \max_{C \in \mathcal{C}(G)} \sum_{e \in C} \alpha_e, \quad (11)$$

where  $\mathcal{C}(G)$  is the registered cycle universe. If  $F(G) > 0$ , every registered cycle sign is forced positive. Equality removes the obstruction only in a minimizing great-semicircle configuration; reaching  $\pi$  is necessary, not sufficient, for a negative sign.

**Corollary 3** (Conservative upper-bound certificate). *If registered upper bounds  $\bar{\alpha}_e$  satisfy*

$$\sum_{e \in C} \bar{\alpha}_e < \pi,$$

*then cycle  $C$  has positive holonomy. A reported positive sign below this bound is a reproducibility check, not evidence of orientability beyond the edge geometry.*

## 2.6 Outer operational gates

Holonomy is still only a geometric diagnostic. A causal claim requires an untouched intervention split showing that loop-sensitive features add held-out predictive value beyond local lineage, rank, path length, Jacobian visibility, and uncertainty. An edit-sectioning claim requires a matched comparison against global and independent edit plans. Figure 1 therefore ends with an outer gate rather than an edit recommendation.

A gauge-invariant block-energy observable,

$$\|B_x^\top h\|_2^2 = h^\top (B_x B_x^\top) h, \quad (12)$$

is independent of the basis within  $E_x$ . A signed component  $e_j^\top B_x^\top h$  is gauge-covariant and needs an explicit transport convention. A rooted spanning tree can make that convention reproducible, but it cannot remove path dependence: every off-tree edge closes an audit loop against the tree path.

## 3 Synthetic instrument validation

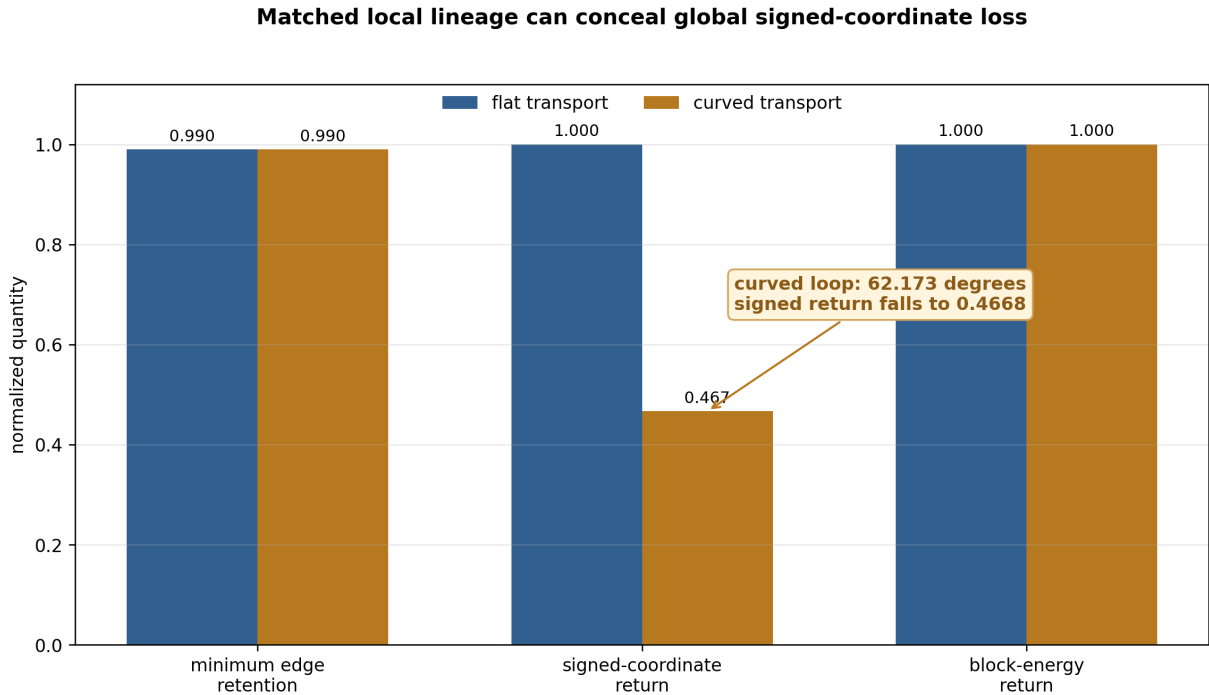
The synthetic program is used to establish identifiability and calibrate the analysis. It does not show that the same objects exist in a transformer.

### 3.1 Matched local lineage can conceal loop inconsistency

A rank-two construction in  $\text{Gr}(2, 4)$  matches the minimum edge worst-direction retention at 0.990033 in flat and curved systems. The curved system has slightly *better* mean edge lineage, yet its loop transport rotates by  $62.173^\circ$  and returns only 0.466809 signed alignment. Block-energy return is exactly one in both arms (Table 1; Figure 2).

**Table 1:** Matched synthetic separation. The local retention statistic cannot identify the orthogonal transport accumulated around the loop.

| Condition | comm.  | min. ret. | mean lin. | angle          | signed   | energy   |
|-----------|--------|-----------|-----------|----------------|----------|----------|
| flat      | 0.0000 | 0.990033  | 0.992571  | $0.000^\circ$  | 1.000000 | 1.000000 |
| curved    | 2.8914 | 0.990033  | 0.994692  | $62.173^\circ$ | 0.466809 | 1.000000 |



**Figure 2:** The flat and curved arms have the same minimum edge retention. Only the signed coordinate records the accumulated loop rotation; gauge-invariant block energy is unchanged.

This is the decisive separation between edge and loop certificates. Singular values determine principal angles but not the polar factor in Equation (2). The synthetic construction shows that this discarded factor is realizable and operationally consequential for signed coordinates.

### 3.2 Curvature and estimator noise have different scaling laws

Estimated subspaces can produce spurious loop rotation even when the true connection is flat. The registered null independently resamples every node of a flat connection in the tangent space. Planted curvature produces phase proportional to enclosed area; independent local estimation errors accumulate as a random walk, producing signed-phase variance proportional to perimeter. The frozen fits are:

The result does not license a fixed raw-angle threshold on transformers. A real loop must be compared

**Table 2:** Synthetic scaling calibration.

| Scaling test                                   | slope    | intercept | $R^2$    |
|------------------------------------------------|----------|-----------|----------|
| curved maximum canonical angle vs. area        | 83.8378  | 0.01178   | 0.999975 |
| flat-noise signed-angle variance vs. perimeter | 0.001078 | 0.00183   | 0.947972 |

with an empirical null conditioned on rank, loop perimeter, edge-retention statistics, and estimator uncertainty. High observed holonomy with a high null quantile is an estimation problem, not a mechanistic discovery.

### 3.3 Planted causal path-transfer calibration

A separate rank-four CPU fixture transports the same root coordinate to one target by two paths. One direct-sum block is curved and an orthogonal block remains flat, providing exposed and low-exposure directions in the same bundle. The downstream response is  $\tanh(Jz)$ , so the relevant Lipschitz bound is exact inside the edit space. Across 3,456 rows and 12 loops:

- the maximum curved exposed displacement is 0.982108 and maximum response disagreement is 1.183766;
- frozen holonomy features reduce grouped held-out squared error by 89.0005%, with a grouped-bootstrap descriptive 90% range [84.9073%, 92.3200%];
- a seeded permuted-holonomy leakage null yields only 5.2462% relative reduction;
- the conservative bound authorizes 91.291% of candidates with zero false authorizations; and
- flat, energy-invariance, gauge, path-identity, and orientation controls all pass.

This establishes that the measurement can recover a *planted* causal path-dependence signal beyond local lineage and Jacobian visibility. It is the shape of a successful outer gate, not transformer evidence.

### 3.4 Holonomy sectioning is conditionally useful

The edit-sectioning control shifts the true edit-identity boundary away from a fixed holonomy boundary. For  $N = 8$  cells,  $J = 2$  patches, patch bias  $B^2$ , noise variance  $\sigma^2$ , and  $n$  observations per cell, the exact expected patch-minus-independent utility is

$$\left(1 - \frac{J}{N}\right) \frac{\sigma^2}{n} - B^2 = 0.75 \frac{\sigma^2}{n} - B^2. \quad (13)$$

At zero offset, the patch plan beats both independent and global edits. In every registered decoupled defect cell, where bias dominates the variance advantage, it loses with the predicted sign. Holonomy patches are therefore not automatically good edit units; their value is a calculable bias–variance tradeoff that depends on whether geometric boundaries match edit identity.

## 4 Transformer evidence: the binding layer is context-conditioned lineage

The real-model sequence is best read as a progression of increasingly discriminating failures and partial recoveries. Table 3 separates evidence classes and prevents retrospective pooling.

**Table 3:** Experiment map. Canonical report paths appear in Appendix A.

| Experiment                   | Main result                                                                                                                                                   | Evidential status                                                                            |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Matched synthetic loop       | Matched minimum edge retention 0.990033; curved transport accumulates $62.173^\circ$ and returns 0.466809 signed alignment while energy stays one.            | Strong synthetic proof that local lineage can miss global signed-coordinate inconsistency.   |
| Synthetic causal transfer    | Frozen holonomy features reduce grouped held-out SSE by 89.0005%; authorization coverage 91.291% with zero false authorizations.                              | CPU-synthetic causal calibration only.                                                       |
| Edit sectioning              | Patch plan wins when geometry and edit identity agree; loses in all registered decoupled defect cells.                                                        | Synthetic falsification of unconditional sectioning value.                                   |
| Qwen0.8B local construction  | No supported rank at any of 512 local nodes; pooling eight contexts recovers 60 of 64 objects.                                                                | Real-model negative: global structure recoverable, local reusable coordinates unsupported.   |
| Qwen0.8B dense-local capture | Sixteen replicas recover one family at common rank one across four layers; context edges control connectivity.                                                | Mixed result: reusable rank-one lineage exists; continuous/non-Abelian holonomy unavailable. |
| Percolation reanalysis       | Every connectivity-critical edge is a context edge; layer 15 has $\tau_{\text{cycle}} = 0.897892$ just below the frozen 0.9 floor.                            | Exact retrospective threshold geometry, not causal usefulness.                               |
| Fresh Stage B                | Fresh layer 23 graph has common rank one, $\beta_1 = 3$ , and coherent positive signs; 4-bit layer 19 lacks a common graph while float16 passes.              | Fresh development-model result, later liveness-qualified.                                    |
| Qwen1.7B functional lineage  | Zero of 12 modules supports a global identity rank; prompt identity explains over 95% of retention variation.                                                 | Independent real-model negative reinforcing the context bottleneck.                          |
| Paired precision case study  | Four-bit precision compresses all eight layer-19 margins, causes one gate crossing, and erodes the layer-19 frustration margin by $23.54^\circ$ or $22.6\%$ . | Strongest real-model holonomy-adjacent result; binary topology never becomes live.           |

#### 4.1 Initial local construction: pooling recovers an object, not local coordinates

The first Qwen0.8B local experiment used two prompts per subcondition, half, and local context. None of 512 node-level rank tests supported any rank from one through eight. This is not a holonomy null; the object failed before a lineage graph could be built. Pooling progressively more contexts changed the support rate:

$$2 \text{ contexts} : 65/512, \quad 4 \text{ contexts} : 373/512, \quad 8 \text{ contexts} : 60/64.$$

The warranted conclusion is diagnostic: a pooled global between-class structure exists on most site/family cells, but the sample-local construction cannot certify reusable coordinates across contexts.

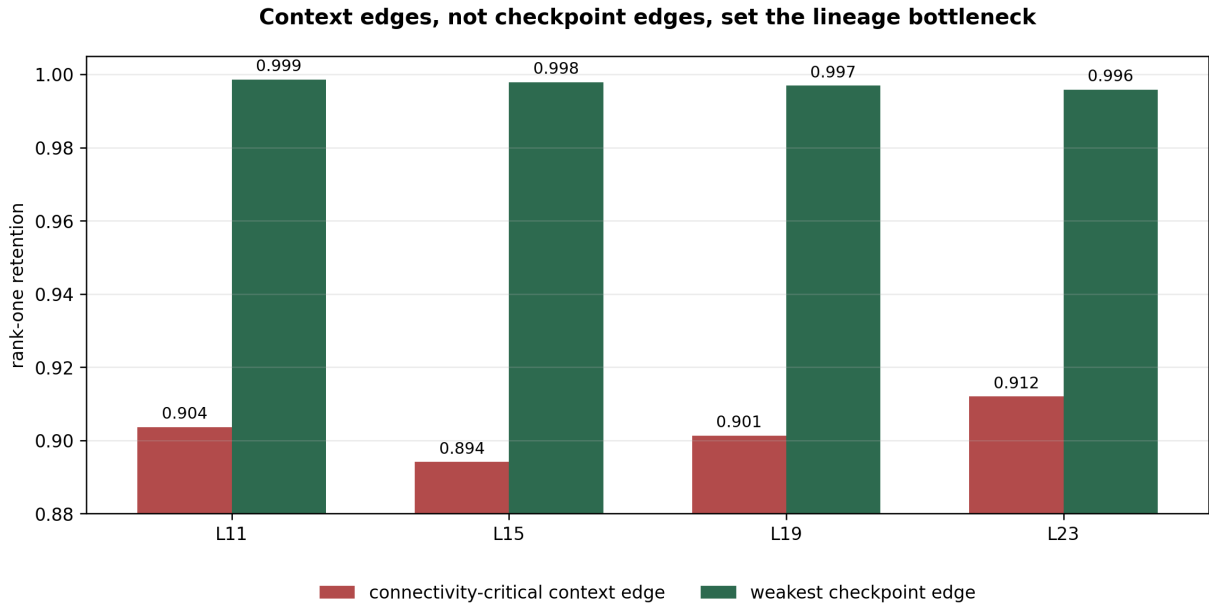
## 4.2 Dense-local recovery reaches rank one, with context as the weak axis

Increasing to 16 replicas per subcondition recovers one behavior family, **graph\_reachability**, at a common rank of one across layers 11, 15, 19, and 23. Individual nodes include four local rank-four cases, but the min-over-context grid rank remains one. At the frozen lineage floor, the admitted graph summaries are:

**Table 4:** Dense-local Qwen0.8B graph-reachability lineage.

| Layer | common rank | admitted edges / 10 | $\beta_1$ | admitted generator loops / 3 |
|-------|-------------|---------------------|-----------|------------------------------|
| 11    | 1           | 10                  | 3         | 3                            |
| 15    | 1           | 4                   | 0         | 0                            |
| 19    | 1           | 9                   | 2         | 2                            |
| 23    | 1           | 10                  | 3         | 3                            |

Checkpoint edges are extremely clean: the weakest checkpoint edge across the four layers is at least 0.9958. Context edges are far weaker, with layerwise minima near 0.894–0.906. The exact percolation reanalysis confirms that every connectivity-critical edge is an adjacent-context edge (Figure 3).



**Figure 3:** Exact retrospective Qwen0.8B dense-local percolation result. The red bars are connectivity-critical context edges; the green bars are the weakest checkpoint edges. Context, not the base-to-naive checkpoint transition, controls graph connectivity in every layer.

This is the first durable real-model positive: one context-conditioned rank-one lineage object exists. It is also a boundary result. A common rank of one makes continuous non-Abelian holonomy unavailable, regardless of the higher ranks seen at isolated nodes.

## 4.3 Percolation reanalysis: unavailable at a floor is not absent

The exact descending filtration locates both connectivity and cycle thresholds:

**Table 5:** Qwen0.8B dense-local exact threshold geometry.

| Layer | $\tau_{\text{conn}}$ | $\tau_{\text{cycle}}$ | $\beta_1$ at 0.9 | critical edge class |
|-------|----------------------|-----------------------|------------------|---------------------|
| L11   | 0.903603             | 0.929846              | 3                | context             |
| L15   | 0.894128             | 0.897892              | 0                | context             |
| L19   | 0.901281             | 0.923886              | 2                | context             |
| L23   | 0.911987             | 0.918675              | 3                | context             |

Layer 15’s earlier `holonomy_unavailable` status is exactly a threshold effect. At  $\tau = 0.9$ ,  $\beta_1 = 0$ ; at  $\tau = 0.897892$ , the first cycle appears; and at zero,  $\beta_1 = 3$ . The correct statement is not “layer 15 has no topology.” It is “no cycle is admitted at the frozen floor.”

A fresh Stage B capture then recovers at layer 23 a common rank-one graph with all ten edges,  $\beta_1 = 3$ , construction/validation agreement, and positive signs on the measured cycle basis. Layer 19’s float16 control also recovers a coherent rank-one graph, while the 4-bit cross-runtime construction lacks a common graph because one context is rank zero. These are fresh sign results, but their interpretation depends on the rank-one liveness theorem.

#### 4.4 Independent Qwen1.7B localization again points to prompt identity

A separate Qwen1.7B functional-lineage study evaluates 12 module families and supports zero global identity ranks. Its frozen failure-localization analysis decomposes balanced retention variation across prompt, replica, checkpoint, and residual/interaction factors:

**Table 6:** Balanced main-effect shares in the Qwen1.7B failure-localization analysis.

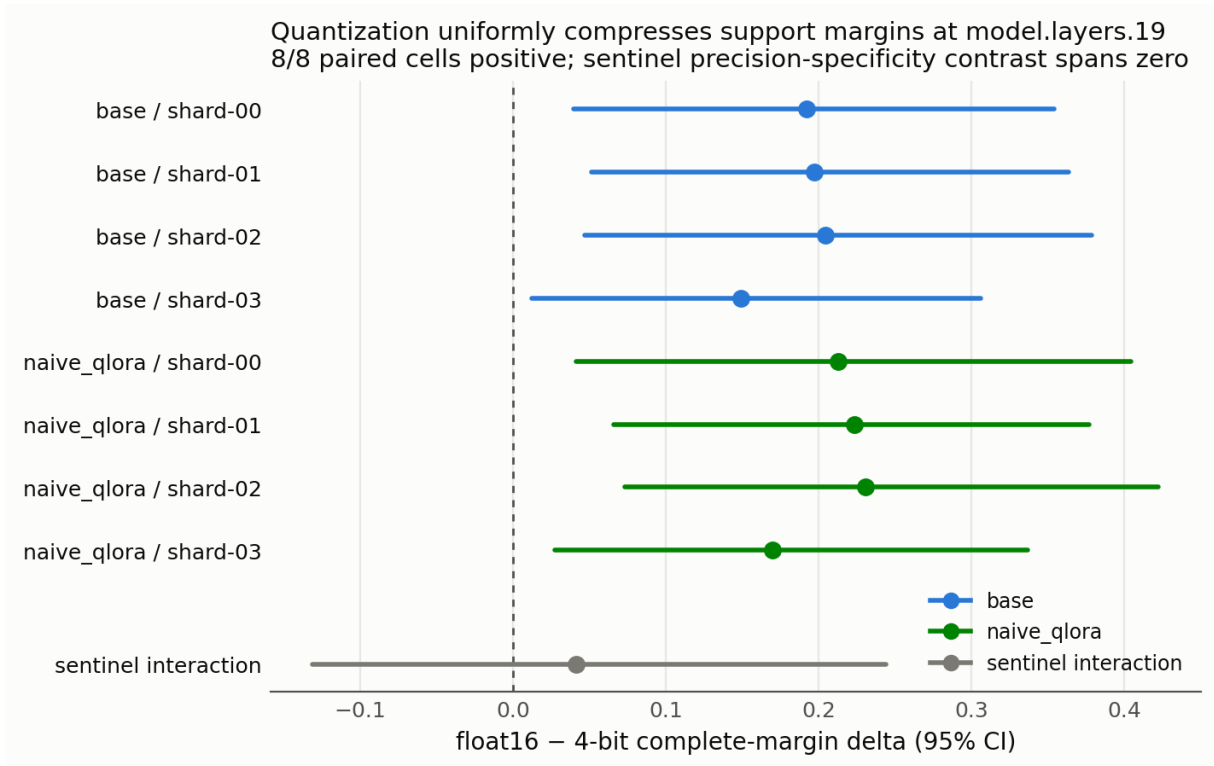
| Layer | prompt  | replica                                                             | checkpoint | residual / interactions |
|-------|---------|---------------------------------------------------------------------|------------|-------------------------|
| 16    | 95.163% | 0.542%                                                              | 0.00014%   | 4.295%                  |
| 22    | 98.152% | remaining share spread across replica, checkpoint, and interactions |            |                         |
| 27    | 98.593% | remaining share spread across replica, checkpoint, and interactions |            |                         |

The exact percentages should not be pooled with the Qwen0.8B lineage graph because the objects and protocols differ. Their qualitative convergence is nevertheless important: prompt or context identity, not checkpoint identity, is the dominant axis along which global reuse fails.

#### 4.5 Paired precision: metric damage precedes live topology

The focused precision experiment crosses two runtime precisions, two weight states, two sites, and four fresh context shards on byte-identical prompts. The low-precision arm is NF4 4-bit double quantization with float16 compute; the comparison arm uses unquantized float16 weights [Dettmers et al., 2023]. All activations are stored in float32. The design is not an fp32 arithmetic comparison.

All eight layer-19 paired cells have positive float16-minus-4-bit support-margin differences, from 0.149 to 0.231 (Figure 4). One 4-bit layer-19 cell crosses its registered local support boundary. A preregistered fine-tuning-specific sentinel fails its specificity rule, so the damage is broad across states and shards rather than unique to the naive QLoRA state. The registered precision shift in cycle-emergence thresholds is not established.

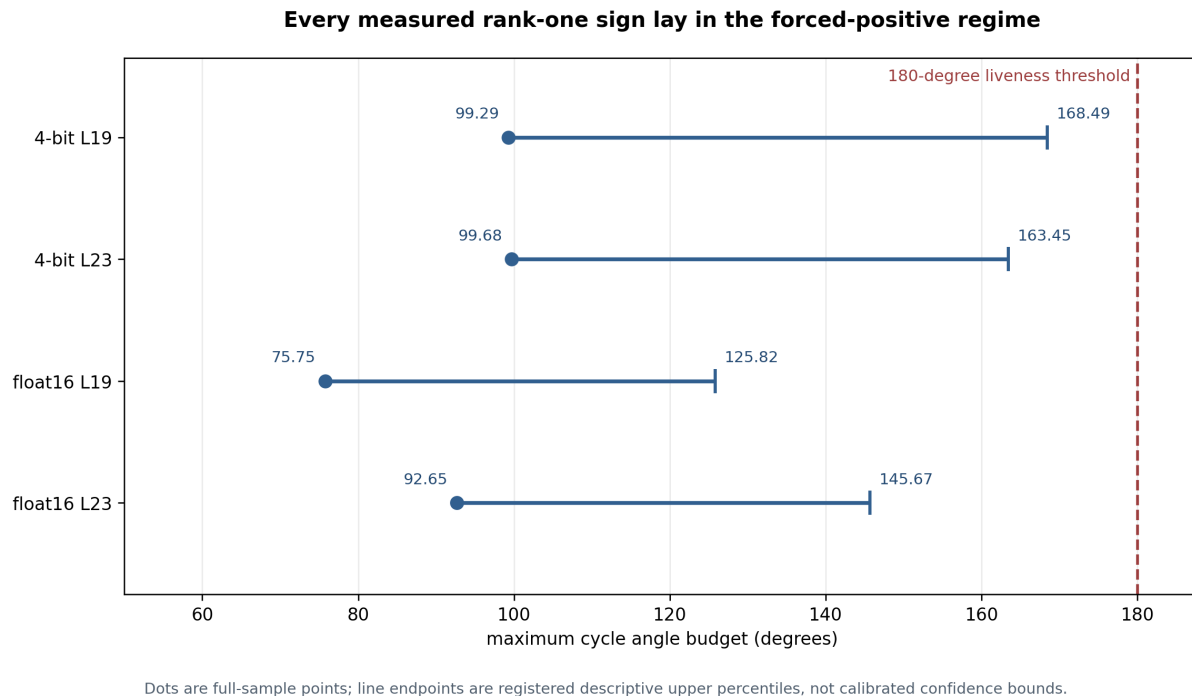


**Figure 4:** Float16-minus-4-bit complete-margin differences at Qwen0.8B layer 19. All eight point differences are positive. Displayed intervals are descriptive and were not multiplicity-corrected; the robust registered statement is the 8-of-8 sign consistency.

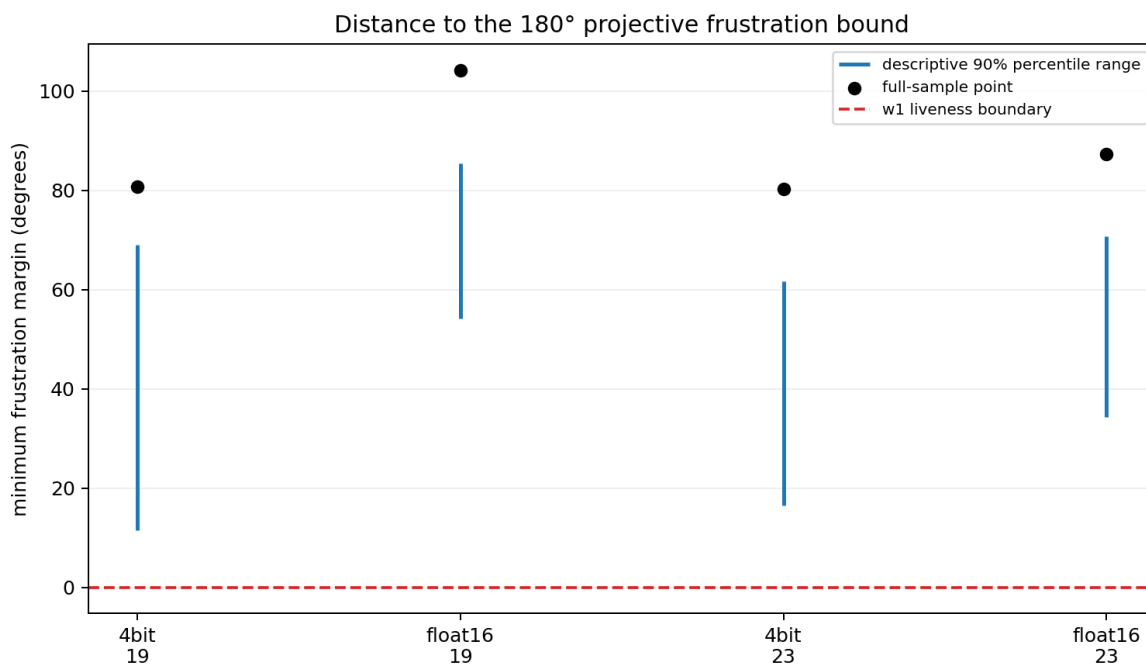
All measured rank-one cycle signs are positive. Theorem 2 shows why that fact is not evidence of topological stability. The maximum point-estimate cycle budget is below  $100^\circ$  in every precision/site graph. Even the registered descriptive upper-percentile endpoints remain below  $180^\circ$  (Figure 5). The binary alternative was geometrically unavailable.

The live observable is the graph frustration margin in Equation (11). At layer 19, 4-bit precision consumes  $23.54^\circ$ , or 22.6%, of the float16 margin. Its registered descriptive 90% range is  $[5.11^\circ, 53.11^\circ]$ , meeting the frozen descriptive decision rule. Layer 23 erosion is  $7.03^\circ$  and does not meet that rule (Figure 6).

The sealed historical phrase `precision_stable Diagnostically_thin` remains part of the audit trail, but its interpretation is superseded. The correct result is: precision erodes support and loop margins, while the observed binary topology is forced by the remaining metric geometry.



**Figure 5:** Rank-one liveness in the paired precision capture. Dots are full-sample maximum cycle budgets; segments end at registered descriptive upper percentiles, not calibrated confidence bounds. Every measured graph remains inside the forced-positive regime.



**Figure 6:** Distance to the 180° rank-one frustration boundary. Positive values certify that all registered cycle signs are forced positive. The vertical ranges are registered descriptive percentile summaries, not calibrated confidence intervals.

## 5 What the sorting pattern does—and does not—mean

The broader mathematical pattern is a hierarchy of *claim obligations*. Table 7 states the scientific question at each layer and the meaning of failure.

**Table 7:** Certificate obligations.

| Layer                     | Question answered                                                                        | Failure means                                             | Failure does not mean                                               |
|---------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------|
| Local support             | Is a registered object distinguishable from its null at this node?                       | Object unavailable under this construction and threshold. | The semantic feature ceased to exist.                               |
| Edge lineage              | Can adjacent supported objects be identified at a common rank?                           | No certified reusable local coordinate across that edge.  | The pooled representation is absent.                                |
| Cycle availability        | Does the admitted graph have nontrivial $H_1$ ?                                          | Holonomy unavailable at the frozen floor.                 | The complete graph has no cycles.                                   |
| Holonomy                  | Does transport compose path-independently?                                               | A loop obstruction or orientation issue may exist.        | The obstruction is above estimator noise or behaviorally important. |
| Liveness / noise          | Could the categorical alternative have occurred, or does the loop exceed a matched null? | Result is forced or noise-dominated.                      | Holonomy mathematics is wrong.                                      |
| Outer causal / sectioning | Does the geometry add value on untouched outcomes?                                       | No edit or control claim.                                 | The geometry is useless as a diagnostic.                            |

In the paired precision case, the observed traversal can be annotated more precisely:

$$\begin{array}{ccccccc}
 \underbrace{\text{support margins compress}}_{\text{measured}} & \longrightarrow & \underbrace{\text{one frozen gate crosses}}_{\text{registered decision}} & \longrightarrow & \underbrace{\text{cycle angle budget rises,}}_{\text{derived from retention}} \\
 & \longrightarrow & \underbrace{\text{frustration margin falls}}_{\text{definition}} & \longrightarrow & \underbrace{\text{positive signs remain forced.}}_{\text{theorem}}
 \end{array}$$

This is a legitimate account of that experiment. It is not a universal physical law. In another system, a local rank can disappear before a margin visibly compresses; a cycle can appear before connectivity; a higher-rank loop can reverse orientation despite high edge retention; or a live loop can remain orientable. The stack orders *what must be certified before stronger language is warranted*.

Three conclusions survive the full sequence.

- S1. Local transfer quality does not certify global signed identity.** The matched synthetic control supplies an exact counterexample.
- S2. Context is the measured weak axis.** Qwen0.8B connectivity thresholds are set by context edges, and Qwen1.7B retention variation is overwhelmingly prompt-conditioned. This is convergent evidence, not a pooled estimate or cross-model law.
- S3. Topological observations need liveness certificates.** A stable discrete class is non-evidential when the measured geometry excludes the alternative. At rank one, the frustration margin is the required companion statistic.

## 6 Operational posture

The current evidence supports a conservative separation between gauge-invariant and signed-coordinate uses.

**Gauge-invariant branch.** Projector or block-energy observables such as Equation (12) remain well-defined under basis rotation. They can be monitored without pretending that one component is globally canonical. This is the defensible branch when the scientific question is subspace occupancy or energy.

**Signed-coordinate branch.** A signed monitor, reward, or edit needs a transport connection, an explicit rooted-tree convention, off-tree loop audits, a common rank, and a live/noise-valid loop universe. The current real-model program has not passed those gates. One global signed edit is therefore unsupported.

**Sectioned branch.** Context-conditioned patches are a plausible constructive route because the dense-local capture contains four nodes with local rank four even though the global common rank is one. The sectioning control warns that patches should be chosen prospectively and evaluated against global and independent plans. A geometry-aligned patch can reduce variance; a misaligned patch introduces bias.

## 7 Next experiment: context-patch liveness factorial

The next experiment should combine the two unresolved questions—context-conditioned rank and topological liveness—rather than repeat another global rank sweep.

### 7.1 Frozen design

1. **Prospective patches.** Construct context-conditioned patches around the four dense-local nodes that previously supported rank four, using fresh prompts and a deterministic, outcome-free patch rule. Do not choose patches after loop signs or behavioral effects are seen.
2. **Longer context ladders.** Increase the number of context shards so that simple cycles can accumulate a potentially live projective angle budget. Seal the cycle universe before measuring signs.
3. **Precision factorial.** Cross float16, registered NF4 4-bit, and an optional controlled 3-bit arm only if numerical and model-function gates pass. Precision is a treatment, not a proxy for model quality.
4. **Cross-fitted objects.** Fit bases on one prompt split, estimate support and lineage on a disjoint geometry split, and reserve a third untouched split for any causal path-transfer outcome.
5. **Matched nulls.** For  $r \geq 2$ , condition the estimator null on rank, perimeter, retention, and sample size. For  $r = 1$ , report exact pointwise angle budgets and registered upper-bound liveness certificates.

### 7.2 Predeclared regimes and gates

For every admitted rank-one cycle, classify:

- **Forced.** A conservative upper angle budget remains below  $180^\circ$ .
- **Live but orientable.** The budget reaches or exceeds  $180^\circ$ , but the observed sign remains positive.
- **Frustrated.** The budget reaches or exceeds  $180^\circ$  and a negative class appears.

The causal stage opens only if all of the following pass: supported local rank, common-rank edge lineage, nonzero cycle rank, exact path-identity checks, matched noise or rank-one liveness, and protocol integrity. The primary causal question is whether loop-sensitive features add grouped held-out predictive value beyond local lineage, rank, path length, Jacobian visibility, precision, and uncertainty. A negative result after these support gates would be meaningfully stronger than the current unavailable outcomes.

This design distinguishes two hypotheses:

- *Diagnostic dormancy*: holonomy is mathematically correct, but the measurable transformer objects remain too local, too rank-thin, or too geometrically short for it to add information.
- *Path-dependent identity failure*: a supported, noise-valid, live loop universe exhibits path-dependent transport that predicts untouched activation, logit, or edit outcomes.

## 8 Limitations and claim boundary

- The synthetic experiments validate instruments on planted geometry; they do not establish corresponding transformer mechanisms.
- The Qwen0.8B and Qwen1.7B studies use different objects, prompt universes, sites, and protocols. Their numbers are not pooled.
- The strongest real-model positive is a narrow rank-one graph-reachability lineage object. No higher-rank transformer bundle has supported continuous, noise-valid non-Abelian holonomy.
- The precision experiment compares NF4 4-bit weights with float16 compute against unquantized float16 weights. It is not an fp32 arithmetic comparison.
- Several max-type bootstrap intervals are registered descriptive sensitivity summaries without calibrated coverage or multiplicity correction.
- A topological or geometric signal is not a semantic identity proof. No downstream deployment threshold, capability criterion, or safety guarantee was measured.
- No causal transformer edit, VPD efficacy, compactification, recursive self-improvement, or RSI claim follows from this paper.

## 9 Reproducibility and audit trail

The rewrite packet contains reports, frozen protocols, compact receipts, method implementations, figures, and the complete focused precision-paper source archive. Its manifest hashes 101 files. Verification of those hashes establishes byte integrity only; it is not an independent rerun of the underlying model inference.

**Table 8:** Publication provenance.

| Item                                        | Value                                                                  |
|---------------------------------------------|------------------------------------------------------------------------|
| Rewrite packet SHA-256                      | 13648b5e685f35fdd56cd1471b2a0d3b47dc4074b758914a76d01c24ec866ced       |
| Source repository commit recorded in packet | 1490351216907dccc264e1ab088b263fe8ff270                                |
| Manifest verification                       | 101 of 101 listed files matched byte size and SHA-256 before rewriting |
| Focused precision source archive SHA-256    | 14317997dd4d80dff47787b0a9a48cdb21df1f8feffd90767fba34ca8b201d25       |
| Public repository                           | <a href="#">MoralityLabAI/RSITopology</a>                              |
| Feature branch                              | <a href="#">spectral-bundle v0.3 bifiltration branch</a>               |
| Focused public report                       | <a href="#">Qwen0.8B precision-geometry report</a>                     |

The canonical report paths used in this synthesis are listed in Appendix A. Earlier interpretations remain preserved when they were sealed, but later corrections govern scientific language. In particular,

positive rank-one signs below the liveness bound are reported as forced, and `holonomy_unavailable` is never silently converted into zero holonomy.

## 10 Conclusion

The week of experiments did not establish a transformer holonomy control mechanism. It did establish a cleaner scientific object. Representation identity across model conditions is not one similarity number; it is a stack of obligations. A supported local object must survive context-conditioned lineage, the graph must support loops, loop transport must be gauge-valid and live above its geometric or estimator null, and any operational use must survive an untouched outer test.

Synthetic controls show why each layer is distinct. The real-model sequence then identifies the present bottleneck: context-conditioned lineage. More replicas recover rank one, not the hoped-for global higher-rank bundle. Checkpoint edges are clean, context edges set connectivity, and an independent model shows the same prompt-conditioned failure localization. Runtime quantization consumes metric support and frustration margin, but the measured rank-one topology never becomes live.

The resulting posture is neither “holonomy works” nor “holonomy fails.” It is sharper: holonomy is a valid downstream audit of coordinate consistency, but its alternative outcome must be available and its estimator must be calibrated before the result is evidential. Until a context-patched, noise-valid, live loop universe passes a causal outer gate, gauge-invariant block energy is defensible and one global signed edit is not.

## A Canonical report paths

| Role                                       | Repository-relative path                                     |
|--------------------------------------------|--------------------------------------------------------------|
| Synthetic matched holonomy and noise null  | <code>reports/monitor_holonomy_insight.md</code>             |
| Synthetic causal path-transfer calibration | <code>reports/holonomy_causal_transfer_v0_1.md</code>        |
| Synthetic edit-sectioning crossover        | <code>docs/EDIT_SECTIONING_V03_RESULTS.md</code>             |
| Initial Qwen0.8B local falsification       | <code>reports/QWEN08_HOLONOMY_FALSIFICATION_V0_1.md</code>   |
| Dense-local Qwen0.8B recovery              | <code>reports/QWEN08_DENSE_LOCAL_HOLONOMY_V0_1.md</code>     |
| Rank-one $w_1$ reanalysis                  | <code>reports/QWEN08_RANK1_W1_REANALYSIS_V0_1.md</code>      |
| Percolation reanalysis                     | <code>reports/QWEN08_PERCOLATION_REANALYSIS_V0_1.md</code>   |
| Fresh context Stage B                      | <code>reports/QWEN08_CONTEXT_STAGEB_V0_1.md</code>           |
| Qwen1.7B functional lineage                | <code>reports/qwen3_1p7b_functional_lineage_v2.md</code>     |
| Qwen1.7B failure localization              | <code>reports/qwen3_1p7b_failure_localization_v0_1.md</code> |
| Focused precision synthesis                | <code>reports/QWEN08_PRECISION_GEOMETRY_PAPER_V0_1.md</code> |

## B Interpretation supersession ledger

| Earlier reading                                                                                   | Current interpretation                                                                                                              |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Stable positive loop signs demonstrate precision-stable topology.                                 | Positive signs were geometrically forced because all measured rank-one angle budgets were below $180^\circ$ .                       |
| More replicas should recover the desired higher-rank global bundle.                               | More replicas recovered one rank-one graph-reachability bundle; disagreement across context shards remained binding.                |
| Layer 15 has no usable topology.                                                                  | Its cycle space appears at $\tau = 0.897892$ , just below the frozen 0.9 lineage floor.                                             |
| A natural-feature orientation-reversal comparison can be pooled with the dense-local line bundle. | The comparator came from a different Qwen1.7B VPD weight-component construction and lacks the same bound cocycle; it is not pooled. |

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